

General rules for writing technical papers

1. Each paragraph should have at least one idea; each paragraph should have at most one idea.
2. The first sentence of a paragraph should describe the content of the paragraph (topic sentence). The body of a paragraph should support the topic sentence. The last sentence of a paragraph should summarize the main points of the paragraph (concluding sentence).
3. By reading the first and last sentence of a paragraph, you should be able to understand the main ideas of the document.
4. One sentence paragraphs are o.k. in technical writing (sometimes).
5. The thought process in a sentence, paragraph, section, and paper should be linear.
6. Always follow the words "this", "those" and "these" by a noun.
7. Try not to use existence sentences (i.e., "it is", "there are", etc.). (e.g., "there are many opportunities for application of this method"). A better way is to write: "many opportunities are available for the application of this method" or "this method has many opportunities for direct application".
8. Try not to use the word "it" in a sentence, as using this word is often ambiguous. The word "it" is often used to simply a compound sentence. For example, "While an NES functions as a damper by absorbing and dissipating energy, it also couples together the vibration modes of the linear structure and results in Targeted Energy Transfer." In other cases, a semicolon is often employed to make things clear. For example, "The second order polynomial track behaves almost linearly as a TMD; thus, it is eliminated from this study due to our desired to have a restoring force with an essential nonlinearity."
9. Read your documents aloud..... you'll find many mistakes that reading silently will not identify.
10. If a sentence can't be read aloud in one breath, then it is too long and should be broken into multiple sentences. {this sentence provides another example of a reasonable use of the word "it".}
11. When you have 3 items in a list, they should be written as: item1, item2, and item3. For two items, no comma is needed: item1 and item2.
12. Use a comma between two complete sentences (i.e., subject, verb, object) before the conjunction: The boy went to the store, and the cat ran across the street. When you have a single subject, no comma is needed: The boy went to the store and bought a bottle of milk.
13. Don't use "in order to", rather, just use "to". For example, rather than say: In order to obtain the results", you should simply write: "To obtain the results".
14. i.e. means "that is"
15. e.g. means "for example"
16. Liberally cite other's work. You can make enemies by not citing other's work.
17. Only use the word "since" in the temporal context. If you are giving a reason for something, use the word "because".
18. Use a hyphen when you use the phrase: "something-based". For example, a "model-based approach to RTHS".
19. Variables should be italicized in equations and in the text. For example, in the term "x-axis", the "x" should be italicized. In the term, "xg", the x is italicized, but the "g" is not, as the "g" stands for "gravity". In the term, "xi", both the "x" and the "i" are italicized, as the "i" is an index. Note that in the expression "mgL/2", the "g" is italicized, because here "g" stands for the gravitational constant.